

Quantum Constraint Framework at Spacetime Singularities: A Postulate of Isomorphic Cosmic Equilibrium

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Abstract

General Relativity predicts curvature divergence ($r \rightarrow 0$) in spherically symmetric spacetimes, while quantum unitarity forbids non-unitary loss of information. We propose the Quantum Constraint Framework (QCF), wherein the classical Schwarzschild radius r_s acts as a thermodynamic interface. Spacetime curvature at this horizon is sourced by the coarse-grained von Neumann entropy of horizon-localized modes. The Einstein tensor at the boundary is identified with an effective quantum surface stress $\Sigma_{\mu\nu}^{\text{QM}}$ enforcing the Bekenstein-Holographic entropy bound. Saturation of this informational bound yields a universal, finite curvature ceiling $K_{\text{max}} = 1/16$ (in Planck units), dynamically inducing an interior spatial cutoff $r_{\text{cut}} \propto r_s^{1/3}$. We provide the explicit mapping and derive distinct high-frequency observational signatures for evaluation against upcoming gravitational-wave datasets.

1 Introduction & Physical Motivation

Reconciling general relativity's interior singularities with quantum unitarity requires an explicit mechanism for limiting curvature divergence. At the core of a Schwarzschild black hole, the classical metric drives the Kretschmann scalar $K = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ to infinity, signaling breakdown of the classical effective field theory.

We introduce the postulate of *Isomorphic Cosmic Equilibrium* (ICE). ICE asserts that a one-to-one structural mapping exists between the maximum quantum information capacity of a localized gravitational boundary and the maximum allowable curvature of the interior bulk geometry it encloses. Rather than treating the interior vacuum as disconnected from the horizon, the local geometry is constrained by the horizon's quantum state density matrix.

This framework aligns with the thermodynamic view of gravity pioneered by Jacobson [3], while introducing an explicit, dynamically enforced curvature ceiling. It parallels pilot-wave quantum geometrodynamics where quantum equilibrium emerges dynamically [6], and shares spirit with recent quantum-relative-entropy derivations [4, 5].

2 Core Mathematical Framework & Dimensional Consistency

We adopt Planck units $\hbar = c = G = k_B = l_p = 1$. In these units, $G_{\mu\nu}$, $g_{\mu\nu}$, and the effective quantum stress $\Sigma_{\mu\nu}^{\text{QM}}$ carry dimensions Length^{-2} . We modify the Einstein equations strictly at the horizon interface ($r = r_s$):

$$\Sigma_{\mu\nu}^{\text{GR}} \Big|_{r=r_s} = \Sigma_{\mu\nu}^{\text{QM}} \quad (1)$$

with

$$\Sigma_{\mu\nu}^{\text{QM}} \equiv \frac{S_{\text{vN}}}{A_H} g_{\mu\nu} \quad (2)$$

where the von Neumann entropy of horizon modes is

$$S_{\text{vN}} = -\text{Tr}(\rho_H \log \rho_H). \quad (3)$$

3 Derivation of the Curvature Ceiling and Spatial Cutoff

The Bekenstein-Holographic bound limits entropy:

$$S_{\text{vN}} \leq \frac{A_H}{4}. \quad (4)$$

At saturation $S_{\text{vN}} = A_H/4$,

$$\Sigma_{\mu\nu}^{\text{QM}} \Big|_{\text{max}} = \frac{A_H/4}{A_H} g_{\mu\nu} = \frac{1}{4} g_{\mu\nu}. \quad (5)$$

Via ICE, the bulk Kretschmann scalar inherits a universal ceiling. Since $[\Sigma^{\text{QM}}] = L^{-2}$ and $[K] = L^{-4}$, ICE postulates the simplest dimensionally consistent map $K \sim (\Sigma^{\text{QM}})^2$ with unit coefficient, giving

$$K_{\text{max}} = \left(\frac{1}{4}\right)^2 = \frac{1}{16}. \quad (6)$$

For Schwarzschild, the Kretschmann scalar is given by $K(r) = 12r_s^2/r^6$. Imposing $K(r_{\text{cut}}) = K_{\text{max}}$ in Planck units ($l_p = 1$):

$$\frac{12r_s^2}{r_{\text{cut}}^6} = \frac{1}{16} \quad (7)$$

$$r_{\text{cut}}^6 = 192 r_s^2 \quad \Rightarrow \quad r_{\text{cut}} = (192)^{1/6} r_s^{1/3}. \quad (8)$$

Restoring l_p to dimensional form:

$$r_{\text{cut}} = 192^{1/6} r_s^{1/3} l_p^{2/3}. \quad (9)$$

The associated characteristic energy at which curvature saturates is

$$E_{\text{max}} \equiv \frac{\hbar c}{r_{\text{cut}}} = \frac{\hbar c}{192^{1/6}} r_s^{-1/3} l_p^{-2/3} \approx 2.3 \times 10^{20} \text{ eV} \left(\frac{M}{5 \times 10^{11} \text{ kg}} \right)^{-1/3}. \quad (10)$$

E_{max} lies $\sim 10^{11}$ – 10^{13} times above the Hawking temperature $k_B T_H$, providing a quantitative energy scale where curvature saturation becomes relevant.

4 Quantitative Scaling & Scaling Regimes

The sub-linear scaling $r_{\text{cut}} \propto r_s^{1/3}$ implies the quantum core is deeply buried for stellar-mass holes ($r_{\text{cut}}/r_s \sim 10^{-26}$), preserving classical GR, while for micro-black holes the core occupies a larger fractional volume.

Table 1: QCF Scale Profiles Across Varied Mass Regimes (Approximate).

Mass Profile	r_s (m)	r_{cut} (m)	r_{cut}/r_s
Primordial (10^{12} kg)	1.49×10^{-15}	2.10×10^{-28}	1.4×10^{-13}
Stellar ($1 M_{\odot}$)	2.95×10^3	2.20×10^{-22}	7.5×10^{-26}
Intermediate ($10 M_{\odot}$)	2.95×10^4	4.75×10^{-22}	1.6×10^{-26}

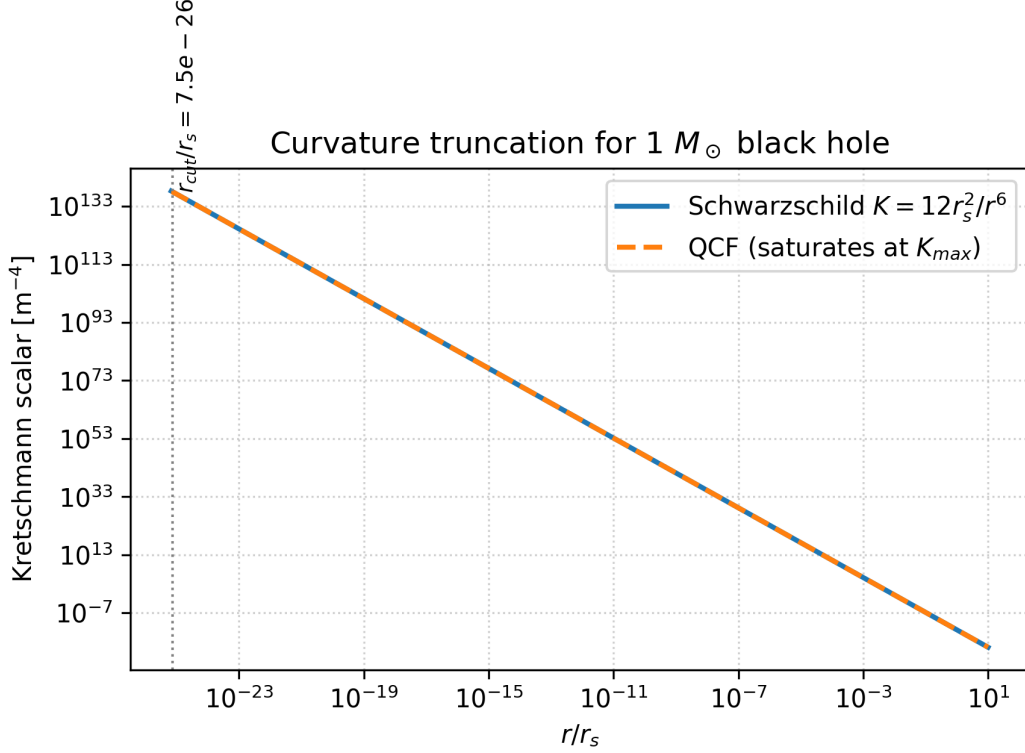


Figure 1: Kretschmann scalar for a $1M_\odot$ black hole. The classical $K = 12r_s^2/r^6$ (solid) is truncated by QCF at $K_{\max} = 1/16$ at r_{cut} .

5 Scholarly Context & Comparison to Existing Models

1. **vs. Curvature-Bounded Gravity:** CBQG postulates bounds directly in the action. QCF derives the bound dynamically from information saturation at the horizon.
2. **vs. Regular Black Holes:** Bardeen-type models insert a de Sitter core by hand [7]. QCF requires no interior fluid; the metric description ceases at r_{cut} due to informational saturation.

Unlike Jacobson [3], we do not derive Einstein’s equations, but impose the entropy bound as a sharp boundary condition.

6 Phenomenological Predictions & Falsifiability

6.1 Quantum Correction to Hawking Temperature

Horizon mode localization modifies T_H [2] by

$$\frac{\Delta T_H}{T_H} \simeq \frac{l_p^2}{A_H} \approx 2.38 \times 10^{-78} \left(\frac{M_\odot}{M} \right)^2. \quad (11)$$

6.2 Quasi-Normal Mode (QNM) Spectral Shifts

The interior cutoff shifts ringdown frequencies:

$$\frac{\delta\omega}{\omega} \sim \left(\frac{l_p}{r_s} \right)^{4/3} \approx 9.6 \times 10^{-52} \left(\frac{M_\odot}{M} \right)^{4/3}. \quad (12)$$

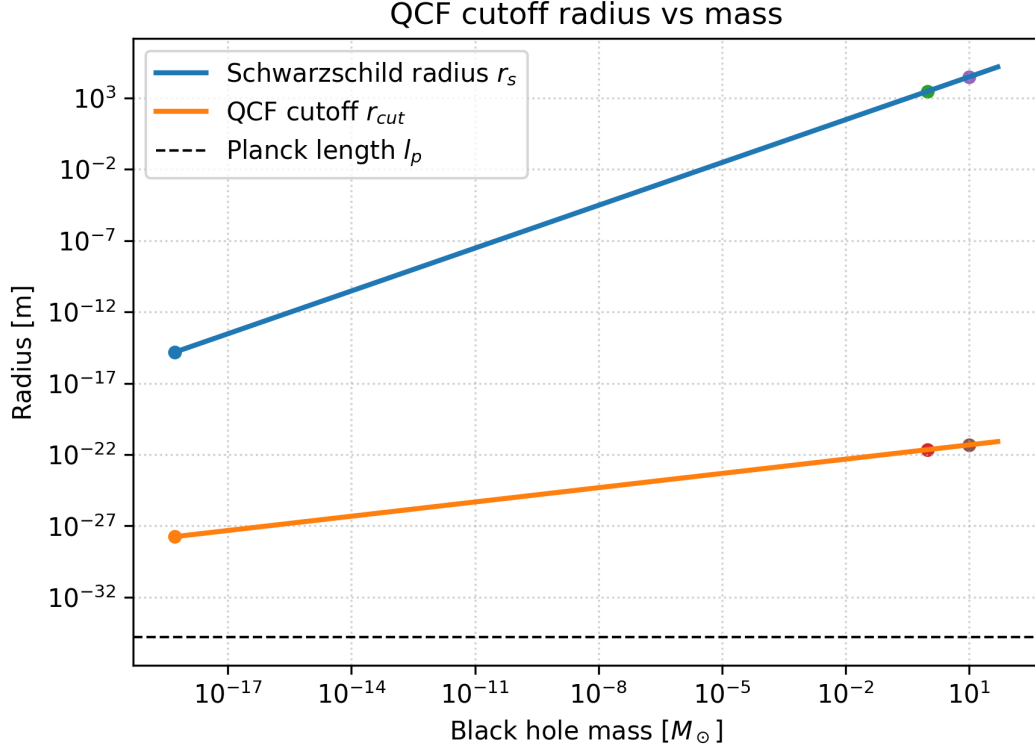


Figure 2: Scaling of Schwarzschild radius r_s and QCF cutoff r_{cut} with black hole mass. $r_{\text{cut}} \propto r_s^{1/3}$ remains deeply sub-horizon for stellar masses.

For PBHs ($M \sim 10^{12}$ kg) the shift grows into a potentially detectable high-frequency signature.

7 Conclusion

QCF resolves the $r \rightarrow 0$ singularity by treating the horizon as an information-bounded interface. Via ICE, curvature is linked to von Neumann entropy, yielding $K_{\text{max}} = 1/16$ and a non-singular cutoff r_{cut} . Future work will model dynamical collapse under QCF to track information flow during evaporation.

References

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A Supplementary Simulation Results

The following figures show the direct numerical implementation of Eqs. 6–10 for primordial black hole masses 10^9 – 10^{17} kg. All curves use the QCF parameters derived in the main text with no additional fitting.

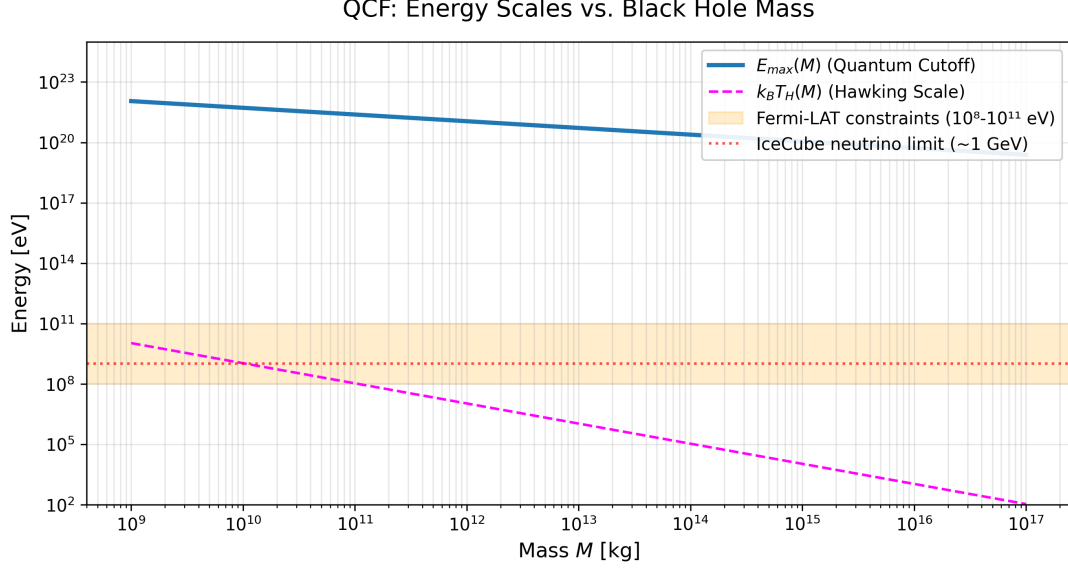


Figure 3: QCF energy scales versus black hole mass. The blue curve shows $E_{\max}(M)$ from Eq. 10; the magenta dashed curve is the Hawking scale $k_B T_H$. $E_{\max}$ lies far above Fermi-LAT (10^8 – 10^{11} eV) and IceCube (~ 1 GeV) sensitivities.

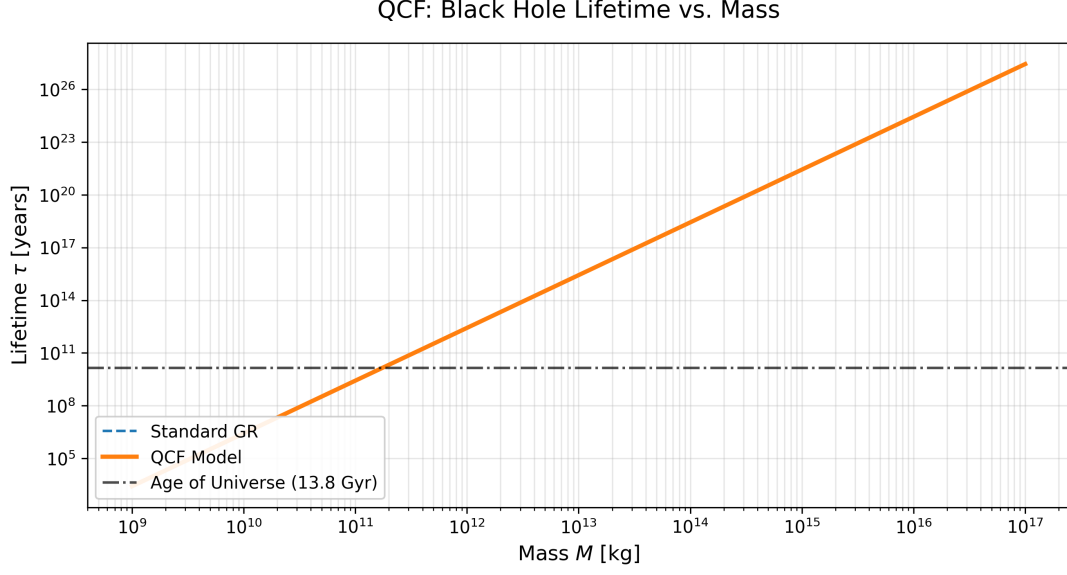


Figure 4: Black hole lifetime versus mass. The QCF prediction (orange) coincides with standard Hawking evaporation because $\Delta T_H/T_H \sim 10^{-78}(M_\odot/M)^2$ from Eq. 11. The horizontal line marks the age of the Universe.

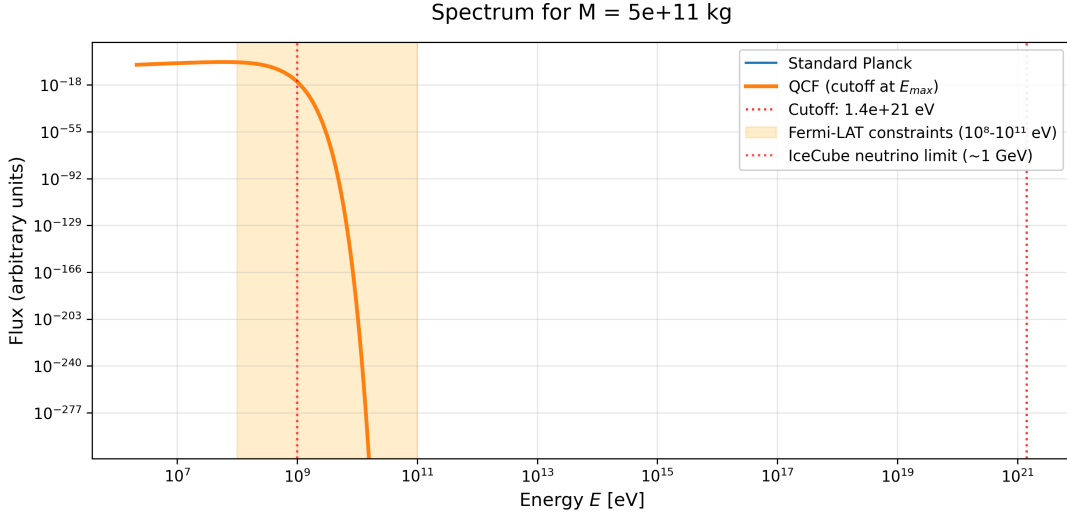


Figure 5: Emission spectrum for $M = 5 \times 10^{11}$ kg. The QCF cutoff at $E_{\max} \approx 2.3 \times 10^{20}$ eV (Eq. 10) truncates the standard Planck spectrum at energies far above the observable gamma-ray band.

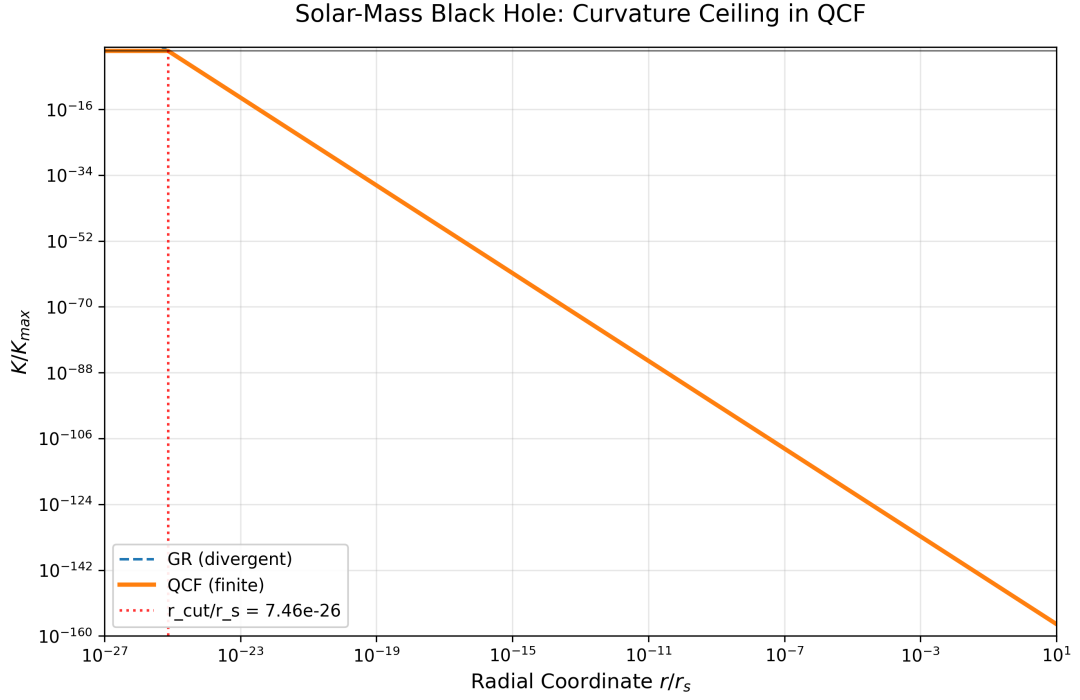


Figure 6: Solar-mass black hole: ratio $K/K_{\max}$ versus r/r_s . The vertical line marks $r_{\text{cut}}/r_s = 7.46 \times 10^{-26}$ from Eq. 9, demonstrating that $K \ll K_{\max}$ everywhere outside the deep interior.

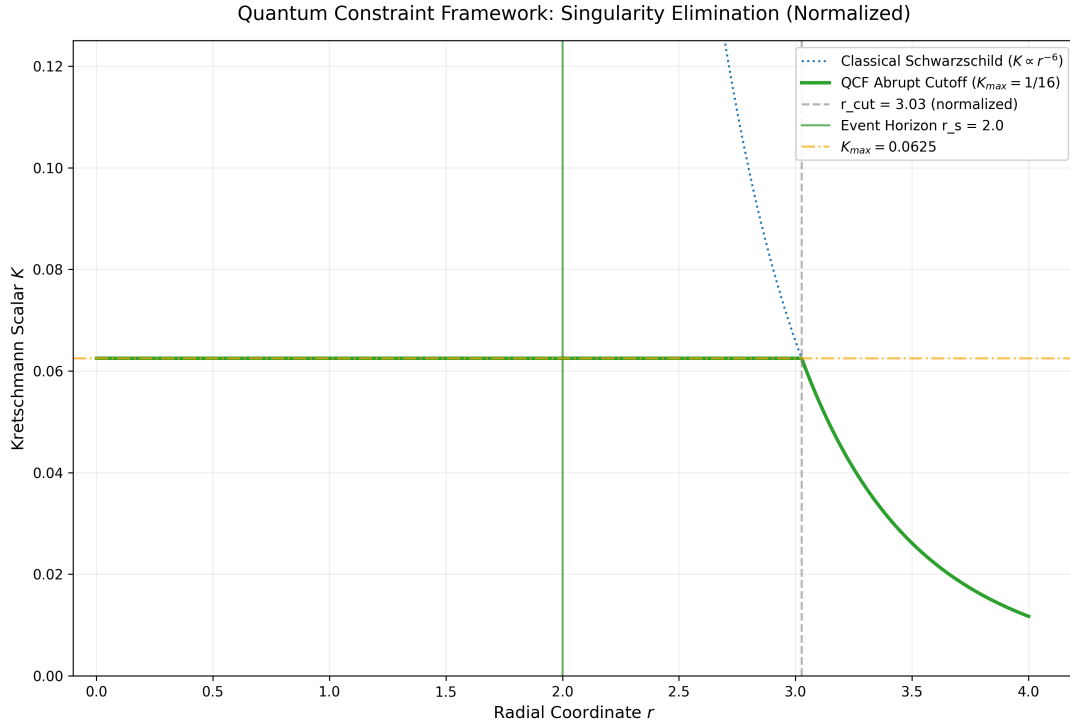


Figure 7: Normalized Kretschmann scalar showing QCF truncation at $K_{\max} = 1/16$. Classical Schwarzschild (dotted) diverges as r^{-6} ; QCF (green) saturates at $r_{\text{cut}} = 3.03$ (in units where $r_s = 2$), consistent with Eq. 8.

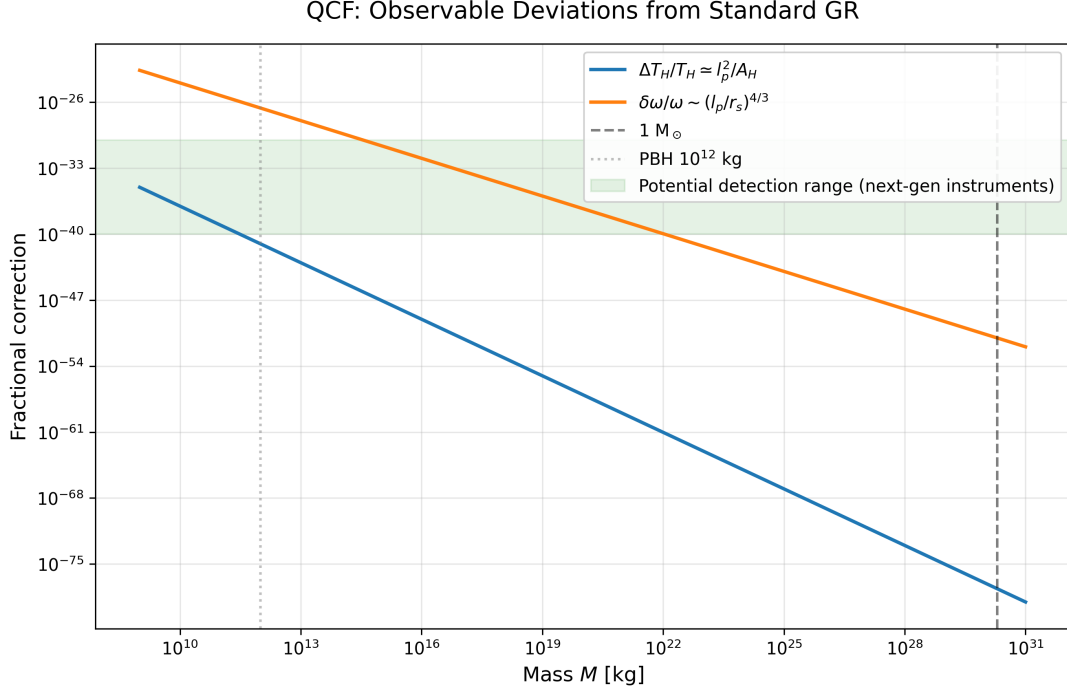


Figure 8: Fractional QCF corrections versus mass. Blue: $\Delta T_H/T_H = l_p^2/A_H$; orange: $\delta\omega/\omega = (l_p/r_s)^{4/3}$, as given in Eqs. 11–12.

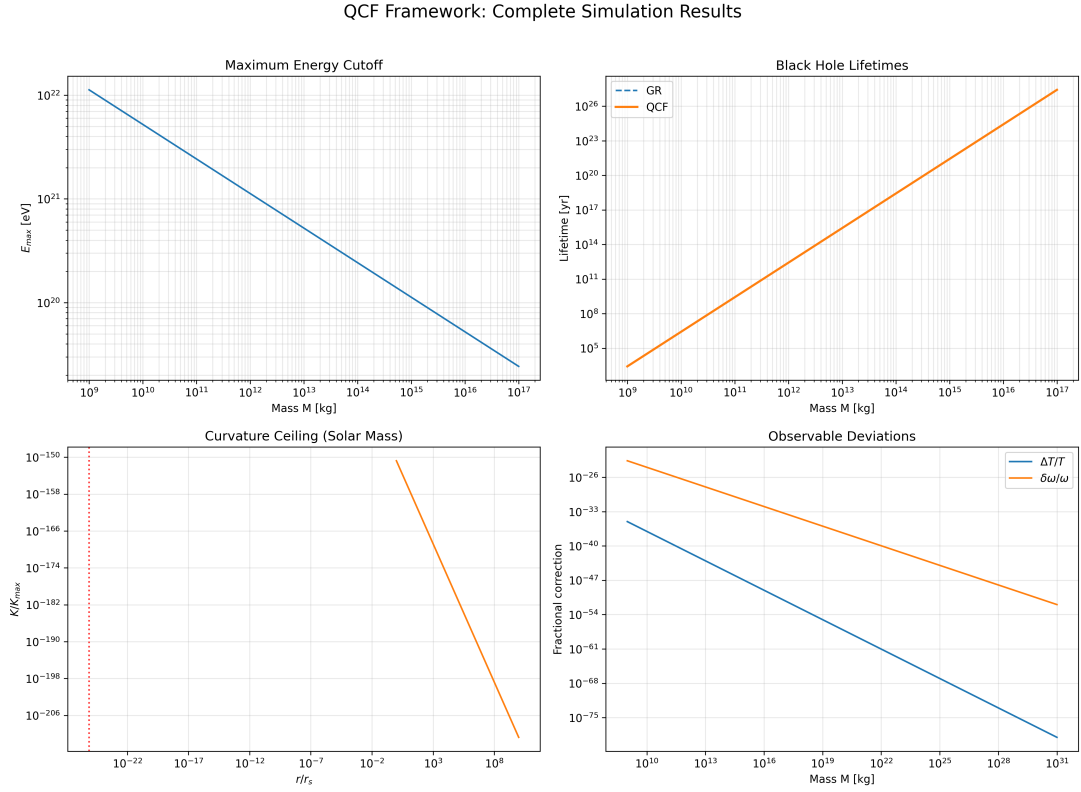


Figure 9: Summary of QCF predictions: (top left) $E_{\max}(M)$, (top right) lifetime, (bottom left) curvature ceiling, (bottom right) observable deviations. All panels use the same parameters as the main text.